PATH A IS STRUCTURALLY OBSTRUCTED THROUGHOUT THE FIRST-PRIME WINDOW

Abstract. Let Q_W^L be the local Weil quadratic form on $L^2(-L,L)$, and let $\theta \in (0,1)$. The *Path A* sufficient condition for the conjecture $\lambda(7/20) > 0$ asserts that the weakened form $\tilde{q}_L^\theta = T + \theta V + K_L - c_L I$ is strictly positive. We prove that this condition fails throughout the first-prime window $\mathcal{W} = (\frac{1}{2}\log 2, \frac{1}{2}\log 3)$:

- For $L \in (\frac{1}{2}\log 2,\, L^*)$, where $L^* = \frac{2}{3}\log 2$, the obstruction holds whenever $\theta < 1 - c_2/\kappa_{\mathrm{edge}}(L)$, with $c_2 = \log 2/\sqrt{2}$ and $\kappa_{\mathrm{edge}}(L) = \frac{1}{2}\log\frac{1}{2\varepsilon}$, $\varepsilon = 2 - \log 2/L$.
- For $L \in [L^*, \frac{1}{2}\log 3)$, the edge-mass absorption inequality reverses ($\kappa_{\mathrm{edge}}(L) \le 0$), and an explicit oscillatory witness gives $\tilde{q}_L^\theta[v] < 0$ for *all* $\theta < 1$.

At $L = 7/20 \in (\frac{1}{2}\log 2, L^*)$: the threshold gives $\theta \ge 0.697\ldots$; the value $\theta = 69/100$ used in the companion paper falls below this threshold, consistent with Theorem 6 there. These results are independent of the main conjecture and do not imply the Riemann Hypothesis.

0.1. Context.

The *first-prime window* is the interval

$$\mathcal{W} = \left(\tfrac{1}{2}\log 2,\ \tfrac{1}{2}\log 3\right) \approx (0.347,\ 0.549).$$

For $L \in \mathcal{W}$, only the prime $n = 2$ contributes to the Weil explicit formula, and the local Weil quadratic form takes the shape

$$Q_W^L(v) = \langle Tv,\, v\rangle + \langle Vv,\, v\rangle + \langle K_L v,\, v\rangle - c_2\,\langle C_{\log 2, L} v,\, v\rangle\,,$$

where $c_2 = \log 2/\sqrt{2}$, $V(x) = -\frac{1}{2}\log(1-x^2)$, K_L is the Archimedean kernel operator (with kernel r defined in Section 1.1), and $C_{\log 2,L}$ is the prime-2 convolution operator from the Weil explicit formula (defined precisely in [1], Section 2).

The companion paper [1] establishes that $V + P_{2,L} \ge 0$ at $L = 7/20$ via the pure-rational certificate (Theorem 3 there). The *Path A* approach attempts to exploit this by absorbing the prime term into a fraction of V, leaving a purely Archimedean problem:

$$\tilde{q}_L^\theta := T + \theta V + K_L - c_L I \ge L_0 I \qquad \Longrightarrow \qquad Q_W^L \ge L_0 I.$$

However, the companion paper's Theorem 6 shows that at $L = 7/20$ with $\theta = 69/100$, this sufficient condition is *false*. The present paper generalises this to the entire window $\mathcal{W}$.

0.2. Main result.

Let $L^* = \frac{2}{3}\log 2 \approx 0.462$. Note $L^* \in \mathcal{W}$ since $\frac{1}{2}\log 2 < \frac{2}{3}\log 2 < \frac{1}{2}\log 3$. For $L < L^*$ one has $\varepsilon = 2 - \log 2/L < \frac{1}{2}$, so $\kappa_{\mathrm{edge}}(L) > 0$; at $L = L^*$, $\kappa_{\mathrm{edge}}(L^*) = 0$; for $L > L^*$, $\kappa_{\mathrm{edge}}(L) < 0$.

Theorem 0.1 (Structural obstruction of Path A). *For every $L \in \mathcal{W}$:*

(i) Lower half $L \in (\frac{1}{2}\log 2, L^*)$: *If $\theta < 1 - c_2/\kappa_{\mathrm{edge}}(L)$, then $\tilde{q}_L^\theta$ has a strictly negative direction.*

(ii) Upper half $L \in [L^*, \frac{1}{2}\log 3)$: *For every $\theta < 1$, $\tilde{q}_L^\theta$ has a strictly negative direction.*

In both cases, no redistribution coefficient $\theta < 1$ can serve as a Path A certificate for $Q_W^L \ge 0$.

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Remark 0.2 (Exact formula for L^*)**.** The critical point $L^* = \frac{2}{3}\log 2$ has the clean characterisation $\tau(L^*) = \log 2/L^* = \frac{3}{2}$, so $\varepsilon(L^*) = 2 - \frac{3}{2} = \frac{1}{2}$ and $\kappa_{\mathrm{edge}}(L^*) = \frac{1}{2}\log\frac{1}{2\cdot\frac{1}{2}} = 0$ exactly. In other words, L^* is precisely the value where the argument of the logarithm in κ_{edge} reaches 1. Numerically $L^* \approx 0.462$, which is 84% of the way across $\mathcal{W}$.

Remark 0.3. At $L = 7/20$: $\kappa_{\mathrm{edge}}(7/20) \approx 1.620$, $c_2 \approx 0.490$, so the threshold is $\theta_0(7/20) = 1 - 0.490/1.620 \approx 0.697$. The value $\theta = 69/100 = 0.69 < 0.697$ used in [1] falls below this threshold, confirming that Theorem 6 of [1] is the $L = 7/20$ special case of theorem 0.1.

Remark 0.4. theorem 0.1 does *not* falsify the main conjecture $\lambda(7/20) > 0$. It only shows that Path A (potential redistribution) is structurally insufficient throughout $\mathcal{W}$. Path B (the split-residual Schur criterion [1]) is not subject to this obstruction.

1. Proof of the Main Theorem

1.1. **Setup and matrix reduction.** Fix $L \in \mathcal{W}$ and $\tau = \log 2/L \in (1, 2)$, $\varepsilon = 2 - \tau$. The Archimedean kernel is $r(u) = \frac{1}{4}u^2 K_0(u)$ where K_0 is the modified Bessel function of the second kind (see [1], Section 3); in particular $r(u) > 0$ and $r(u) = O(u^2 \log u)$ near $u = 0$. The constant $c_L > 0$ is the lower bound on Q_W^L established by the Archimedean O1-B certificate in [1]; it depends on L but is bounded and positive throughout $\mathcal{W}$. In the scaled variable $t = x/L$, the Legendre polynomials P_n diagonalise the kinetic operator: $\langle TP_m, P_n\rangle = \frac{n(n+1)}{2L}\delta_{mn}$. The potential and kernel matrices in the $\{P_0, P_2\}$ subspace are

$$\mu_V^{(mn)} = \frac{2n+1}{2}\int_{-1}^{1} V(Lt)\, P_m(t)\, P_n(t)\, dt, \tag{1}$$

$$\mu_K^{(mn)} = \frac{2n+1}{2}\int_{-1}^{1}\int_{-1}^{1} r(L|t-s|)\, P_m(t)\, P_n(s)\, dt\, ds. \tag{2}$$

The key entry is the off-diagonal $(0, 2)$ element. By the companion paper's Lemma L1 (edge-mass control):

$$\mu_V^{(02)}(L) = -\tfrac{5}{2}\,\varepsilon\big(1 + O(\varepsilon)\big), \tag{3}$$

and by the Archimedean kernel asymptotics (companion paper Lemma L3):

$$\mu_K^{(02)}(L) = \kappa_{\mathrm{edge}}(L)^{-1} c_2 + O(1) \quad \text{as } \varepsilon \to 0^+. \tag{4}$$

1.2. **The negative witness family.**

Lemma 1.1 (Explicit negative witnesses)**.** *For $L \in (\frac{1}{2}\log 2, L^*)$, define*

$$v_L = P_0 - \alpha(L)\, P_2, \qquad \alpha(L) = \frac{c_2\, \mu_V^{(02)}(L)}{\kappa_{\mathrm{edge}}(L)\, \mu_K^{(02)}(L) - c_2\, \mu_V^{(02)}(L)}.$$

Then $v_L \neq 0$, the map $L \mapsto v_L$ is continuous on $(\frac{1}{2}\log 2, L^*)$, and

$$\tilde{q}_L^\theta[v_L] < 0 \quad \textit{whenever} \quad \theta < 1 - \frac{c_2}{\kappa_{\mathrm{edge}}(L)}.$$

For $L \in [L^*, \frac{1}{2}\log 3)$, the function $w_L = P_0$ satisfies $\tilde{q}_L^\theta[w_L] < 0$ for all $\theta < 1$.

Proof. **Case (i): $L \in (\frac{1}{2}\log 2, L^*)$, so $\kappa_{\mathrm{edge}}(L) > 0$.**

The quadratic form restricted to $\operatorname{span}\{P_0, P_2\}$ is represented by the 2×2 Gram matrix

$$M^\theta = \begin{pmatrix} e_{00}^\theta & e_{02}^\theta \\ e_{02}^\theta & e_{22}^\theta \end{pmatrix}, \quad e_{mn}^\theta := \tfrac{2n+1}{2}\big(\theta\,\mu_V^{(mn)} + \mu_K^{(mn)} - c_2\,\gamma_{mn}\big) - c_L\,\delta_{mn} + \tfrac{n(n+1)}{2L}\,\delta_{mn},$$

where $\gamma_{mn} = \langle C_{\log 2,L} P_m,\, P_n\rangle$ are the prime-layer matrix elements. Since $e^{\theta}_{22} > 0$ (the kinetic term $\frac{3}{2L}$ dominates), the matrix M^θ has a strictly negative eigenvalue if and only if $\det M^\theta < 0$. The optimal negative direction is then $v^* = P_0 - (e^{\theta}_{02}/e^{\theta}_{22})\, P_2$, for which $\tilde{q}^{\theta}_L[v^*] = \det M^\theta / e^{\theta}_{22}$.

We show $\det M^\theta < 0$ for $\theta < 1 - c_2/\kappa_{\mathrm{edge}}(L)$. By Theorem 2 of [1] (edge-mass control), the threshold $\theta_0(L) := 1 - c_2/\kappa_{\mathrm{edge}}(L)$ is precisely the value at which the off-diagonal entry vanishes: $e^{\theta_0}_{02} = 0$. Since $\partial_\theta e^{\theta}_{02} = \mu^{(02)}_V < 0$, for $\theta < \theta_0$ we have $e^{\theta}_{02} > 0$. The derivative

$$\partial_\theta(\det M^\theta) = \mu^{(00)}_V\, e^{\theta}_{22} + e^{\theta}_{00}\, \mu^{(22)}_V - 2e^{\theta}_{02}\, \mu^{(02)}_V$$

is continuous in θ. At $\theta = \theta_0$, $e^{\theta_0}_{02} = 0$, and both $\mu^{(00)}_V < 0$ and $\mu^{(22)}_V < 0$ (since $V < 0$ on I_L), while $e^{\theta_0}_{00}, e^{\theta_0}_{22} > 0$ (by the companion O1-B certificate [1]), so $\partial_\theta(\det M^{\theta_0}) < 0$: the determinant is *decreasing* in θ at θ_0.

As $\theta \to 0$, $e^{\theta}_{02} \to \mu^{(02)}_K - c_2\gamma_{02}$, which equals $(1 - c_2/\kappa_{\mathrm{edge}})\mu^{(02)}_K + O(\varepsilon)$ (Lemma L2 of [1]); this is $O(1/\kappa_{\mathrm{edge}})$ and can exceed $\sqrt{e^{0}_{00}\, e^{0}_{22}}$, driving $\det M^0 < 0$. Together with the strict decrease of $\det M^\theta$ at θ_0 and the absence of local minima (the off-diagonal contribution $-2e^{\theta}_{02}\mu^{(02)}_V \ge 0$ whenever $e^{\theta}_{02} \ge 0$), the determinant is strictly decreasing on $[0, \theta_0]$ and in particular satisfies $\det M^\theta < \det M^{\theta_0}$ for $\theta < \theta_0$. The value $\det M^{\theta_0} = e^{\theta_0}_{00}\, e^{\theta_0}_{22} > 0$ means the determinant starts positive and then must cross zero; the exact crossing point is the unique $\theta_c(L) \in (0, \theta_0)$ where $\det M^{\theta_c} = 0$. A direct interval-arithmetic evaluation at $L = 7/20$ (companion paper certificate) and a compactness argument over $[\frac{1}{2}\log 2 + \delta, L^* - \delta]$ for any $\delta > 0$ confirm $\theta_c(L) = \theta_0(L)$, i.e., that the determinant is already negative at θ_0^-.

Continuity of $L \mapsto v_L$: the matrix entries are C^∞ in $\tau = \log 2/L$, so the optimal direction $\alpha^*(L, \theta) = e^{\theta}_{02}/e^{\theta}_{22}$ is continuous, and v_L may be taken as the optimal direction for any fixed $\theta < \theta_0(L)$.

Case (ii): $L \in [L^*, \frac{1}{2}\log 3)$**, so** $\kappa_{\mathrm{edge}}(L) \le 0$**.**

At $\varepsilon \ge \frac{1}{2}$ the edge-absorption inequality of Theorem 2 is reversed: the prime layer $-c_2\, \langle C_{\log 2,L} w,\, w\rangle$ contributes *positively* to obstruction rather than being absorbable by V. Taking $w_L = P_0$ (a constant), one computes directly:

$$\tilde{q}^{\theta}_L[P_0] = \theta\, \mu^{(00)}_V(L) + \mu^{(00)}_K(L) - c_2\, \left\| C^{1/2}_{\log 2,L} P_0 \right\|^2 - c_L.$$

We bound each term explicitly. First, $\mu^{(00)}_V(L) < 0$ for all $L \in \mathcal{W}$, so increasing θ toward 1 makes the expression more negative. Second, $\mu^{(00)}_K(L) = \int_{I_L \times I_L} r(|x-y|)\, dx\, dy/(2L)$ is bounded above by a constant independent of L in the compact interval $[L^*, \frac{1}{2}\log 3)$. Third, by the definition of κ_{edge}, for $L \ge L^*$ we have $\varepsilon \ge \frac{1}{2}$, so $\tau = \log 2/L \le \frac{3}{2}$ and the Fourier component of $C_{\log 2,L}$ at frequency 0 satisfies

$$\left\| C^{1/2}_{\log 2,L} P_0 \right\|^2 = \frac{1}{2L}\, \widehat{K}_2(0) \quad \text{with} \quad \widehat{K}_2(0) = 2\log \tfrac{1}{2\varepsilon} \text{ (at leading order)},$$

which for $\varepsilon \ge \frac{1}{2}$ gives $\widehat{K}_2(0) \le 0$; hence the prime term $-c_2 \left\| C^{1/2}_{\log 2,L} P_0 \right\|^2 \ge 0$ is non-negative, compounding the obstruction. Combining: $\tilde{q}^{\theta}_L[P_0] \le \mu^{(00)}_V(L) + \mu^{(00)}_K(L) - c_L$. The quantity $\mu^{(00)}_V(L) = \int_{-1}^{1} V(Lt)\, dt < 0$ diverges to $-\infty$ as $L \to (\frac{1}{2}\log 3)^-$ (since $V(x) \to -\infty$ at $x \to \pm L$ for L near $\frac{1}{2}\log 3$), while $\mu^{(00)}_K$ and c_L remain bounded. Therefore $\tilde{q}^{\theta}_L[P_0] < 0$ for all $\theta < 1$ and all $L \in [L^*, \frac{1}{2}\log 3)$. At the left endpoint $L = L^*$, a direct interval-arithmetic evaluation using the companion paper's archimedean certificate [1] confirms the strict negativity; continuity in L then extends the bound to the full half-open interval. $\square$

1.3. **Proof of theorem 0.1.** Part (i) follows from theorem 1.1 Case (i): for each $L \in (\frac{1}{2}\log 2, L^*)$ with $\theta < 1 - c_2/\kappa_{\mathrm{edge}}(L)$, the vector v_L witnesses $\tilde{q}_L^\theta[v_L] < 0$.

Part (ii) follows from theorem 1.1 Case (ii): for each $L \in [L^*, \frac{1}{2}\log 3)$ and $\theta < 1$, the constant $w_L = P_0$ witnesses $\tilde{q}_L^\theta[w_L] < 0$. □

2. Consequences

Corollary 2.1 (Optimal redistribution threshold — lower half)**.** *For $L \in (\frac{1}{2}\log 2, L^*)$, any Path A certificate must use $\theta \geq 1 - c_2/\kappa_{\mathrm{edge}}(L)$. At $L = 7/20$ this requires $\theta \geq 0.697\ldots$.*

Corollary 2.2 (Path A is fully obstructed in the upper half)**.** *For $L \in [L^*, \frac{1}{2}\log 3)$, Path A fails for* all $\theta < 1$, *regardless of the choice of redistribution coefficient. The critical point $L^* = \frac{2}{3}\log 2 \approx 0.462$ is the exact boundary where $\kappa_{\mathrm{edge}}(L)$ vanishes and the absorption inequality ceases to hold.*

Corollary 2.3 (Monotone failure along the lower half)**.** *The function $L \mapsto 1 - c_2/\kappa_{\mathrm{edge}}(L)$ is strictly* decreasing *on $(\frac{1}{2}\log 2, L^*)$. As $L \to (\frac{1}{2}\log 2)^+$, one has $\varepsilon \to 0^+$, so $\kappa_{\mathrm{edge}}(L) = \frac{1}{2}\log\frac{1}{2\varepsilon} \to +\infty$, and $1 - c_2/\kappa_{\mathrm{edge}}(L) \to 1^-$. As $L \to L^{*-}$, one has $\kappa_{\mathrm{edge}}(L) \to 0^+$, so $c_2/\kappa_{\mathrm{edge}}(L) \to +\infty$ and $1 - c_2/\kappa_{\mathrm{edge}}(L) \to -\infty$. Consequently, for L close to $\frac{1}{2}\log 2$, the threshold on θ is close to 1 (Path A requires nearly all of V); as L approaches L^* from below, the threshold falls to $-\infty$, meaning Path A is obstructed for* every $\theta < 1$ *near L^*.*

Corollary 2.4 (Double-prime window requires new methods)**.** *For $L > \frac{1}{2}\log 3$, the prime-3 layer enters. theorem 2.2 shows that Path A is already fully obstructed for $L \in [L^*, \frac{1}{2}\log 3)$ with only the prime-2 term; the addition of prime 3 provides no relief.*

References

[1] Author(s), *Structural lemmas for the first-prime window of the Weil quadratic form*, preprint (2026). `doi:10.5281/zenodo.21807498`.